COM3023

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

COMPUTER SCIENCE

Examination, May 2024

Machine Learning and AI

Module Leader: Dr Zeyu Fu

Duration: TWO HOURS

Answer ALL questions. No word count specified.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

Candidates are permitted to use *non-programmable* portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

Question 1

(a) What is the main difference between a non-informative prior, a weakly-informative prior and an informative prior using normal distribution? Explain which kind of prior choice would result in the largest entropy.

(5 marks)

(b) Given a simple linear regression model, $y_i = wx_i + \epsilon$, for $i = (1, \ldots, n)$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$, derive w and σ^2 via maximum likelihood estimation. Hint: an example Gaussian distribution is given as, $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$.

(10 marks)

(c) Can the product of likelihood and prior be used to make probabilistic statements? Explain the answer.

(2 marks)

(d) What is the main use of Bayesian information criterion?

(2 marks)

(Total 19 marks)

Question 2

(a) Is the Naïve Bayes classifier a Bayesian method? Explain your answer. What is the meaning of 'naïve' in Naive Bayes classifier?

(4 marks)

(b) Explain the key differences between Laplace approximation, variational approximation and Markov Chain Monte Carlo approximation.

(5 marks)

(c) The distribution of a random variable θ is given by:

$$p(\theta|m, n, j) = \theta^{m+1}(\theta + 1 + n)^j,$$

where m, n, j are constants. Find the Laplace approximation of the distribution shown above for $m = 4, n = 6, j = 20$. Give your answers up to two decimal places.

(9 marks)

(d) For logistic regression, explain why a logarithmic function is not a suitable choice for the link function?

(3 marks)

(Total 21 marks)

Question 3

(a) What is the difference between reverse and forward Kullback–Leibler divergence? State two methods to perform optimisation for variational approximation.

(5 marks)

(b) Show that the maximum entropy distribution over a finite (discrete) set $\{m, m+1, \ldots, n-1, n\}$ with $m < n$ is a uniform distribution. (Hint: use Lagrange multipliers)

(10 marks)

(c) What is the key difference between importance sampling and rejection sampling? State one limitation of these two sampling methods.

(5 marks)

(Total 20 marks)

Question 4

A hidden Markov model can be represented as: $\lambda = (S, O, A, B, \pi)$, where $S = (s_1, \ldots, s_n)$ are the states, $O = (o_1, \ldots, o_t)$ are the observations, A is the transition probability matrix, B is the emission probability matrix and π is the initial probability vector.

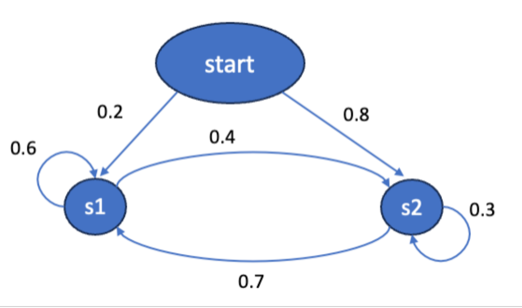

Figure 1: A partially drawn diagram of a hidden Markov model in which s1 and s2 are states.

(a) For the model shown in Figure 1, what are the initial probability vector π and the transition probability matrix A?

(4 marks)

(b) Complete the above transition diagram given the emission probability matrix $B = \begin{bmatrix} 0.7 & 0.3 \\ 0.9 & 0.1 \end{bmatrix}$.

(4 marks)

(c) For the observation sequence $(o_1 \rightarrow o_2)$, use the Viterbi algorithm to estimate the likelihood of the hidden Markov model in the previous question. Show your calculations at each step of the Viterbi algorithm. Give your answers up to four decimal places.

(10 marks)

(Total 18 marks)

Question 5

(a) Explain the meaning of the five key terms in reinforcement learning: *action*, *state*, *reward*, *policy* and *value*.

(5 marks)

(b) Consider a k-armed bandit problem with $k = 3$ actions denoted as *x*, *y*, *z*. A sequence of actions and rewards are given in the following table:

t	Action	Reward
1	x	2
2	x	2
3	z	3
4	y	1
5	x	2
6	z	3

Table 1: A sequence of actions and rewards.

(i) Given that the initial estimates of values, Q_0, for all actions is zero, i.e., $Q_0(a) = 0$ for all a, estimate $Q_t(a)$ for all actions using the sample-average technique and find the optimal actions at each time step, where $t = 1, 2, 3, 4, 5, 6$.

(12 marks)

(ii) Was an ϵ-greedy action selection applied in the above problem? Explain your answer.

(5 marks)

(Total 22 marks)